

WHAT IS UAV VS DRONE:

A UAV IS A UNMANNED AERIAL VEHICLE. IT IS A COMPONENT IN UNMANNED AIRCRAFT SYSTEM THAT IS USE COCKPIT OR HUMAN. CODES TO TAKEOFF AND LAND ARE PREPROGRAMMED INSIDE FLIGHTCONTROLLER WHICH IS A COMPONENT IN UAV. DRONE IS A COMMON NAME FOR AN UNMANNED AIRCRAFT. UAV OR DRONE CAN BE ACCESSED BY REMOTE CONTROL THROUGH HUMANS.UAV IS A TECHNICAL TERM USED IN ENGINEERING.WHILE DRONE IS A COMMON NAME IN PUBLIC FOR UNMANNED AIRCRAFT.EVERY UAV IS A DRONE, BUT EVERY DRONE IS NOT UAV.

USES OF UAV:

UAV ARE USED IN:

1. AGRICULTURE
2. SURVEILLANCE
3. BUILDING SHOOTS
4. SCIENTIFIC RESEARCH
5. ARCHEOLOGICAL RESEARCH
6. FUNCTION SHOOTS
7. LOGOSTICS AND TRANSPORT
8. MEDIA
9. REAL ESTATE
- 10.MILITARY DEFENCE AND ATTACK

TYPES OF DRONES:

THERE ARE MAIN FOUR TYPES OF DRONES.MOSTLY THEY CONSIST OF DIFFERENT PROPELLER.THEY ARE:

MULTIROTOR DRONES:

THESE DRONES HAVE MORE THAN FOUR PROPELLER. ITS IS MAINLY USED IN AGRICULTURAL FIELD, PHOTOGRAPHY AND LOGISTICS.

FIXEDWING DRONE:

IT IS LIKE AIRPLANE. IT MOVES IN FORWARD MOTION TO GENERATE LIFT. IT HAS TWO PROPELLER.

SINGLEROTOR DRONES:

IT HAS ONLY SINGLE PROPELLER. AND IT HAS HIGH BATTERY LIFE. WHICH MEANS LESS PROPELLER MORE FLIGHT TIME. IT CAN ALSO MOVE FORWARD AND BACKWARD.

HYBRID(VTOL) DRONES:

IT HAS HIGH FLIGHT TIMES THAN MULTIROTOR DRONES AND SINGLE ROTOR DRONES. IT IS USED IN MULTIARY SURVIELLANCE, EMERGENCY MEDICAL DELIVERY AND AGRICULTURAL SURVEYING.

BASIC COMPONENTS OF DRONES:

1.FRAME:

THE BODY OF THE DRONE. HOLDS ALL COMPONENTS TOGETHER.

2.FLIGHT CONTROLLER(FC):

IT IS THE BRAIN OF THE COMPUTER. IT CONTROLS THE DRONE MOVEMENT AND PROCESS THE SENSORS IN THE DRONE.

3.MOTORS AND PROPELLER:

MAINLY BLDC MOTORS ARE USED. ITS MAIN FUNCTION IS TO SPIN THE PROPELLER TO GENERATE LIFT. IT CONVERTS ROTATIONAL ENERGY OF MOTOR INTO THRUST. MAKES DRONE FLY TURN AND HOVER.

4.ELECTRONIC SPEED CONTROLLERS:

IT CONTROLS MOTOR SPEED. AND RECEIVE COMMANDS FROM FLIGHT CONTROLLER.

5.BATTERY:

IT PROVIDES ELECTRICAL POWER TO DRONE.

6.POWER DISTRIBUTION BOARD:

DISTRIBUTES POWER TO THE DRONE.

7.RECEIVER AND TRANSMITTER:

RECEIVER RECIEVS COMMAND FROM THE TRANSMITTER. TRANSMITTER IS THE REMOTE CONTROL USED TO OPERATE DRONE.

8.SENSORS:

HELP WITH NAVIGATION AND SAFETY. EXAMPLE
LiDAR, OBSTACLE AVOIDANCE SENSORS.

9.IMU:

CONTAINS ACCELEROMETER AND GYROSCOPES. HELPS
TO TILT AND MOVEMENT.

10.GPS:

PROVIDES LOCATION OF DRONE.

1.WHAT IS PIDCONTROL:

IT HELPS IN DRONE STABILITY AND
BALANCED WHILE FLYING.P - PROPORTIONAL AND
CORRECTS THE CURRENT ERROR.
I - INTEGRAL, IF DRONE SLOWLY DRIFTS TO ON SIDE. IT
CORRECTS THE DRIFT .D- DERIVATIVE. PREDICTS THE
FUTURE ERROR AND REDUCES SHOOTING.

2.WHAT IS MAVLINK:

IT IS A COMMUNICATION PROTOCOL TO
EXCHANGE DATA BETWEEN DIFFERENT COMPONENTS.
SIMPLY IT CONNECTS LAPTOP AND DRONE AND SHARES
MESSAGE.

3.AUTOPILOT FIRMWARE:

IT IS SOFTWARE PROGRAM INSTALLED INSIDE
FLIGHTCONTROLLER THAT CONTROLS DRONE.

INSTALLED PYTHON VERSION 3.11.4

CA Select Command Prompt

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dell>PYTHON --VERSION
unknown option --VERSION
usage: PYTHON [option] ... [-c cmd | -m mod | file | -] [arg] ...
Try `python -h' for more information.

C:\Users\dell>python --version
Python 3.11.4

C:\Users\dell>
```

INSTALLED DRONKIT AND PYMAVLINK:

```
C:\Users\dell>pip install dronekit
Requirement already satisfied: dronekit in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.9.2)
Requirement already satisfied: pymavlink>=2.2.20 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from dronekit) (2.4.49)
Requirement already satisfied: monotonic>=1.3 in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from dronekit) (1.6)
Requirement already satisfied: lxml in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from pymavlink>=2.2.20->dronekit) (6.1.1)
Requirement already satisfied: fastcrc in .\AppData\Local\Programs\Python\Python311\Lib\site-packages (from pymavlink>=2.2.20->dronekit) (0.3.6)
```

INSTALLED MISSION PLANNER :



INSTALLED SITL:



FIRST DRONE CONNECTION SCRIPT:

CODE:

```
from dronekit import connect

print("Connecting to vehicle...")
vehicle = connect('tcp:127.0.0.1:5760', wait_ready=True)
print("connection successfull")
vehicle.close()
print("connection closed")
```

OUTPUT:

```
===== RESTART: C:\Users\dell\Documents\test.py =====
Connecting to vehicle...
CRITICAL:autopilot:APM:Copter V3.3 (d6053245)
CRITICAL:autopilot:Frame: QUAD
connection successfull
connection closed
...
```

IN THIS PYTHON CODE IDLE PYTHON 3.10.8 64 BIT IS INSTALLED, DRONEKIT, SITL THROUGH COMMAND PROMPT AND MISSION PLANNER IS INSTALLED. AND FIRST CONNECTION PROGRAM OF DRONE IS SUCCESSFUL.

BASIC ARM AND TAKEOFF:

CODE:

```
*test.py - C:\Users\deli\Documents\test.py (3.10.8)
File Edit Format Run Options Window Help
from dronekit import connect, VehicleMode
import time

print("Connecting to vehicle...")
vehicle = connect('tcp:127.0.0.1:5760', wait_ready=True)

def arm_and_takeoff(aTargetAltitude):
    print("Basic checks")
    while not vehicle.is_armable:
        print("Waiting for vehicle to initialise...")
        time.sleep(1)

    print("Arming motors")
    vehicle.mode = VehicleMode("GUIDED")
    vehicle.armed = True

    while not vehicle.armed:
        print("Waiting for arming...")
        time.sleep(1)

    print("Taking off!")
    vehicle.simple_takeoff(aTargetAltitude)

    while True:
        altitude = vehicle.location.global_relative_frame.alt
        print("Altitude: %.2f" % altitude)
        if altitude >= aTargetAltitude * 0.95:
```

OUTPUT:

```
Connecting to vehicle...
CRITICAL:autopilot:APM:Copter V3.3 (d6053245)
CRITICAL:autopilot:Frame: QUAD
CRITICAL:autopilot:Calibrating barometer
CRITICAL:autopilot:Initialising APM...
CRITICAL:autopilot:barometer calibration complete
CRITICAL:autopilot:GROUND START
Basic checks
Waiting for vehicle to initialise...
Waiting for vehicle to initialise...
Waiting for vehicle to initialise...
Waiting for vehicle to initialise...
Waiting for vehicle to initialise...
Waiting for vehicle to initialise...
Arming motors
Waiting for arming...
Waiting for arming...
Waiting for arming...
Waiting for arming...
ERROR:autopilot:ARMING MOTORS
CRITICAL:autopilot:GROUND START
Waiting for arming...
WARNING:dronekit:Link timeout, no heartbeat in last 5 seconds
CRITICAL:autopilot:Initialising APM...
Taking off!
Altitude: 0.00
Altitude: 0.00
Altitude: 0.00
```


IN THE ABOVE PYTHON SCRIPT THE DRONE ARM AND TAKEOFF. IT CAN BE DONE THROUGH CONNECTING SITL WITH PYTHON PROGRAM. AND THUS ARMING AND TAKE OFF IS SUCCESSFUL..

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